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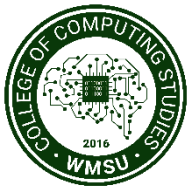
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HULI-KAMOTE: CAMERA-BASED MULTI-VIOLATION DETECTION AND NON-CONTACT APPREHENSION SYSTEM FOR URBAN TRAFFIC ENFORCEMENT IN ZAMBOANGA CITY

A Thesis Presented to the Faculty of

Department of Computer Science

College of Computing Studies

In Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Computer Science

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Approval Sheet

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Acknowledgment

<This section recognizes the persons and organizations who assisted the researchers in the completion of the thesis. Acknowledgments should be expressed simply and tactfully.>

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CHAPTER I

INTRODUCTION

Background of the Study

In recent years, road safety in Zamboanga City has reached a critical point. According to reports from the Philippine Information Agency, traffic-related accidents are rising because of increased vehicle usage and a widespread culture where people do not follow traffic laws. Physical checkpoints and on-the-ground enforcers are commonly employed by local authorities, but this manual operation is very limited due to unpredictable weather, lack of personnel, and the physical risks that traffic operators on busy roads confront, leaving large stretches of the city with little to no enforcement and riders free to ignore the rules.

Despite local ordinances like the strict "No Helmet, No Travel" policy under City Ordinance No. 248, also known as the Revised Traffic Code for Zamboanga City, and Republic Act No. 10054, also known as the Motorcycle Helmet Act, a significant number of riders still travel daily without headgear, often carrying multiple passengers, and this pattern of ignoring the rules does not stop at helmets. This reckless behavior is where the name "kamote riders" comes from, used locally to describe motorists who consistently ignore road rules in ways that put others at risk. For how many months, a Facebook page called "Zamboanga Idiot Drivers" became a regular topic of conversation because it was being shared everywhere. It went viral from actively posting pictures of motorcycle riders caught

breaking rules on Zamboanga roads. What was alarming aside from the violations caused was that concerns about privacy were raised by the people who saw some of the posts that contained recognizable pictures of the people involved without their knowledge. The same reckless pattern is not limited to motorcycle riders. Car drivers contribute to the same problem, with violations such as going against the flow of traffic, illegal parking, and illegal turning, among others.

Deploying an untested system on an actual road was not something we could responsibly do within a thesis timeline. Our adviser recommended keeping the study within a simulation setup so we could verify that the detection logic works before any live deployment is considered. The system is built so that a live camera can be connected in the future, but all testing for this study was done using pre-recorded footage and controlled video input only.

To address this gap, this study proposes a local offline dashboard that automatically flags detected violations, logs the violation type, source camera, confidence score, and a captured frame for each event. Also, the system emails a notification when a violation is recorded. It records the accumulated violations per vehicle by using a demerit point counter and it provides an option for motorists to file an appeal to a separate portal. The system runs on local hardware and does not connect to any government enforcement database. No record it produces is submitted to any agency or used in actual enforcement.

Statement of the Problem

Traffic violations in Zamboanga City are not always recorded consistently because manual monitoring depends on the presence, location, and availability of traffic personnel. As a result, violations that occur outside an enforcer's presence go unrecorded, and the kamote behavior behind them goes unaddressed and keeps going. Without an automated way to detect, notify, and document these violations, there is also no basis for tracking repeat offenders or giving motorists a structured way to dispute a flag.

Specifically, this study aims to answer the following questions:

1. What traffic and driving violations can the system detect from video input?
2. How does the system notify, record, and allow review of detected violations?
3. How accurately does the system log and organize detected violations?
4. What are the limits of the system's performance under simulation conditions?

Objectives

The general objective of this study is to develop HULI-KAMOTE, a prototype designed with an architecture to support live camera implementation for detecting multiple traffic and driving violations. The system will extract license plate numbers, transmit email notifications, process user appeals, track demerit points, and log the results into a local dashboard. The actual testing and evaluation of the

prototype are strictly limited to simulated environments and controlled video footage.

Specifically, this study aims to:

- To gather and prepare a video dataset covering both traffic and driving violations for motorcycles and cars;
- To build and train a detection model that catches multiple violation types at once and follows how a vehicle moves across frames, with each detection tagged to the specific camera that captured it;
- To integrate violation flagging, plate number reading, email notification, user appeal function, and demerit point tracking into a two-interface dashboard with an Admin view for LTO personnel and a Violator Portal for motorists; and
- To evaluate detection performance and logging accuracy under simulation conditions, and gather feedback from LTO personnel as the target users through a live demonstration.

Scope and Limitations

The system built for this study is a prototype that processes traffic video by identifying selected violations, reading the plate number, assigning a violation case number, and storing the results in a local dashboard. The dashboard has two sides, an Admin Dashboard for LTO personnel and traffic enforcers to review flagged violations, and a Violator Portal where a registered motorist can check their

records and file an appeal. When a violation gets logged, the system sends an email notification automatically, and accumulated violations per plate number are tracked through a demerit point counter inside the dashboard. Rather than testing on a live road, this study keeps all testing within simulation, recorded footage, and controlled video input. This architecture is built to support a live camera connection in the future, but that is not what this study tests.

The target users of this system are LTO personnel and local traffic enforcers in Zamboanga City. They are the ones expected to use the Admin Dashboard and serve as evaluators when the system is demonstrated.

Two vehicle types are covered, specifically motorcycles and cars. The violations selected between motorcycles and cars are almost the same: counterflow driving, illegal parking, illegal turn, disregarding traffic signs, and no visible plate number. Additional violations for motorcycles are no helmet, missing side mirror, and overloading. The panel also requested the inclusion of signal light violations, red light running, Swerving, No seatbelt, and pedestrian lane obstruction. These are being considered as extended targets, but whether they make it into the final prototype depends on what the hardware can handle during development. In selecting the primary categories, we considered whether the violation was visible in a single frame or whether it could be inferred from how a vehicle moves across a short sequence of frames. Each detected violation is also tagged to the specific camera that captured it.

For the video material, we will pull from different sources depending on what is available. Pre-recorded traffic footage will be formally requested from barangay offices in Zamboanga City, starting with Barangay Baliwasan, and other barangays will also be approached if that request is delayed or if more footage is needed. Traffic footage from the Metropolitan Manila Development Authority (MMDA) will also be requested through the electronic Freedom of Information (eFOI) Portal. GTA V gameplay recordings will be used as a synthetic source since they produce controlled road scenarios without capturing real people. Public datasets from COCO, Kaggle, and Roboflow Universe will supplement these. If none of the above becomes available, the researchers will record the footage themselves. The target is at least 3,000 images, annotated through Roboflow with an 80/10/10 training split.

Only what is necessary to log the violation will be stored. Faces and personal information of the drivers are not included in the records. The system is tested offline using local computing resources. It does not connect to any government enforcement database or make final enforcement decisions. Email notifications, appeal records, and demerit points are all contained within the prototype and carry no legal weight. Nothing the system logs is used in official enforcement.

The study has several limitations. Because no camera was deployed on an actual public road, the system's output reflects only what occurs under the selected simulation and controlled-footage conditions. Video quality also

affects detection accuracy, particularly when footage is blurry, dark, highly compressed, small in the frame, angled, or blocked. Reading the plate number becomes especially difficult under those same conditions, or when the plate itself is dirty or damaged. Side mirror detection is also harder than the others because mirrors are small objects and the rider's arms often block them in the frame. Movement-based violations, such as counterflow, also require enough frames to show the direction of travel, so a very brief wrong-way movement may not be detected. Because some footage may come from outside Zamboanga City, including GTA V scenes and MMDA recordings, the results may not fully reflect traffic patterns specific to the city.

Significance of the Study

Camera-based traffic detection is not new, but a locally-built and simulation-tested version with documented results is something that did not exist for the City of Zamboanga before this study. The groups listed below each have a different reason to find it useful, ranging from local traffic offices looking at what detection could look like in practice, to future researchers who need a working local example to build from.

For LTO personnel and related agencies, LTO staff are the target users of this system and will take part in the evaluation once the prototype is completed. They will review the system through a live demonstration and their feedback will be documented using the FURPS model and System Usability Scale. What they

say during the evaluation will be recorded and may be referenced in future discussions about traffic monitoring, though any actual use may still require legal and policy clearance from other agencies. If this system is ever implemented, personnel would not need to be in the field at all times and the workload becomes more manageable through the dashboard.

For Zamboanga traffic enforcers, when there is no enforcer present because of bad weather, shortage of manpower, or roads that cannot be covered all at once, the system serves as a surveillance tool that can record and monitor in their place. It is not meant to replace them, it just fills in the gaps they cannot physically be at.

For motorists and commuters, they are the ones that share the road daily, they will be the direct beneficiaries. Although it does not get that far yet, but showing that automated detection is technically buildable here using available local tools is the first step. If kamote behavior gets caught more consistently someday because of systems like this, roads become safer and traffic moves better for everyone using them.

For the College of Computing Studies community, the presented research becomes a reference for future students who want to work on traffic monitoring in Zamboanga City. This documents the entire setup, from dashboard design to detection logic, so the next team working on something similar will not have to start from nothing.

For future researchers, the full workflow including the detection logic, tracking thresholds, dashboard design, email notification, appeal feature, and demerit system is all built out here as a starting point for Philippine traffic detection research. The existing documentation can be further developed upon by anyone wishing to expand this work.

Definition of Terms

Some of the terms below have broader meanings outside this study. What follows is how each one is defined for the purposes of HULI-KAMOTE and its testing setup.

Table 1: Definition of Terms

Term	Definition
1. Admin Dashboard	The LTO staff-facing interface where detected violations are reviewed, managed, and monitored.
2. Bounding Box	A rectangular marker drawn by the detection model around a detected object, such as a vehicle, rider, helmet, plate, or traffic sign.
3. Captured Frame	The selected video frame saved by the system to show the visible basis of a detected violation for reviewer checking.
4. Computer Vision	A field of computing that allows software to process images or video and identify objects, patterns, or actions.
5. Counterflow	A movement-based violation where a vehicle travels against the expected flow of traffic in a lane or road segment.
6. Dashboard	The local web interface where the system displays detected violations, evidence frames, timestamps, vehicle data, and review status.
7. DeepSORT	A tracking method used to follow the same detected vehicle across multiple frames and maintain its assigned track identity.
8. Demerit Point System	A dashboard feature that accumulates violation counts per vehicle plate number.

Term	Definition
9. EasyOCR	An optical character recognition tool used in this study to attempt reading visible license plate characters from cropped plate images.
10. eFOI Portal	A Philippine government website where anyone can file formal requests for public documents online, where MMDA footage request is submitted.
11. False Negative	A case where the system fails to detect a violation that is actually present in the video.
12. False Positive	A case where the system marks a violation even though the violation is not actually present.
13. FastAPI	A Python-based backend framework running on localhost that connects the detection pipeline to the Flutter dashboard.
14. Flutter	The frontend framework used to build the HULI-KAMOTE dashboard, including both the Admin Dashboard and Violator Portal views.
15. Ground Truth	The manually checked reference label used to compare whether the system output is correct during evaluation.
16. HULI-KAMOTE	The prototype developed in this study for detecting selected traffic violations and organizing them in a dashboard.
17. Kamote	A commonly used local term for a driver who consistently ignores traffic rules, practices, and puts others at risk.
18. License Plate Recognition	The process of locating and reading the characters on a vehicle plate from an image or video frame.
19. Live Camera Input	A camera that feeds video directly into the system as it happens, in real time. The architecture of HULI-KAMOTE supports this kind of input, but all testing in

Term	Definition
	this study used pre-recorded footage and controlled video only.
20. Logging and Organization Accuracy Rate	An integration metric that checks whether the system correctly records the violation type, vehicle identity, plate result, and evidence frame for a ground-truth event.
21. mAP@50	mAP@50 is the score we used to see how well the detection model was doing overall. A detection only counts as a hit if the box the model drew overlaps with the actual object by at least 50%.
22. MMDA	The Metropolitan Manila Development Authority manages traffic and urban development in Metro Manila. Traffic footage from MMDA was one of the sources requested for use in this study.
23. OCR	Optical character recognition, or the conversion of visible text in an image into machine-readable characters.
24. Simulation-Based Testing	Testing performed using controlled video, generated scenes, or replayed footage instead of live deployment on public roads.
25. SMTP Server	The email protocol used to send automated violation notifications to registered motorists when a violation is logged.
26. Track ID	A temporary identity assigned by the tracking module to a vehicle while it appears in the video sequence.
27. Traffic Violation	A selected road-rule violation included in the scope of this prototype, such as missing helmet, illegal parking, counterflow, or no visible plate.
28. Violator Portal	The motorist-facing dashboard view where a registered user can check their violation records and file an appeal.

Term	Definition
29. YOLO	An object detection model family used to locate and classify objects in images or video frames.

CHAPTER II

REVIEW OF RELATED LITERATURE

Related Studies

Motorcycle Violations

Upon categorizing the studies to their distinct themes, we found that motorcycle violations are far more difficult to be detected in the camera due to its conditions and sizes. Motorcycles are much smaller than cars that's why cameras may fail to capture them as they can just hide behind a car or vehicles that are bigger than them. That's why most systems that are proposed may either perform good or struggle depending on the models they used.

One particular study by Nishant et al. [5] where they created an AI system that checks if riders are not wearing helmets and added a feature where it can also detect if the said motorcycle does not have side mirrors. Their study focuses more on capturing the motorcycle violation of not wearing helmets when riding. The side mirror detection was only a built-in feature but clearly contributed to the rarely proposed studies of detecting side mirrors in the category of motorcycle violations. Perhaps their study is the most relevant because they combined both detecting the absence of side mirrors and helmets. Unfortunately though, the system has a weakness of finding the side mirrors because mirrors are small as we all know, have different shapes, and are easily hidden behind the driver. Knowing this, side mirrors detection could be one of the challenges we may experience because

aside from the difference of sizes and shapes of motors here in the Philippines, the system we will build will also struggle if we use datasets that are not tested locally.

When locally speaking though, there is a one study that talks about detecting riders that do not have helmets and not wear it properly here in Makati City, where they created an AI-model using a tool called YOLOv5 [6]. Their system groups riders into four categories: First is wearing a full-face helmet, second is wearing a half-face helmet, third is wearing a wrong or invalid helmet like a bicycle cap, and lastly, not wearing a helmet at all. When they categorise the riders into their category, they can achieve a good performance from the model. However though, the system can make mistakes and mix up the groups because the data they used was not equal, meaning some type of helmet pictures had way more examples than the others.

Aside from the challenge of side mirror detection, there is also the challenge of detecting objects at night especially at low-light angle. Two studies have talked about this challenge where the first study is from Bose et al. [7] where they added a built-in feature of fixing the dark video clips before it detects to their system which is an AI-system that was created with a tool called YOLOv8 to detect three types of two-wheelers traffic violations. The type of violations their system only detect are the riders that are not wearing helmets, triple riding, and having a dangerous driving tricks. Most of which are also proposed to our study, excluding the dangerous driving tricks. The only downside of their system is that it can still make

mistakes and miss violations if the motorcycle riders are wearing caps or unusual headwear, as the AI tends to get confused and may mistake the items as actual helmets.

The second study is from Deshpande et al. [8] which is different from all the studies mentioned previously because they built a two-stage system in the context of smart Indian city, meaning it will detect the violation of the motorcycle first and then read the vehicle's license plate number. To achieve this, they used a framework called DetectNet_v2 to first find the motorcycle riders, and then used the YOLOv8 model combined with OCR to read the plate number in real-time. Now here is the problem, their system can still experience missed detections under low-light or blocked conditions and also struggles to localize the riders accurately because when many motorcyclists are crowded very close to one another, their system's tracking boxes may overlap.

From all the studies reviewed in the category of motorcycle violations, most of them feels incomplete compared to what we are building. Most of them fail to address the very purpose of why we are building a multiple violation detection system in the first place. The reason why they are incomplete because most of the studies only focus on detecting the helmetless riding of the motorcycles and their plate number as their only scope or there are some that attempt for the detection of side mirrors but often showed inaccurate results. No study has explored the detection of motorcycles violating the illegal parking, illegal turns, counter flow, excess passengers and disregarding the traffic signs.

Car Violations

Car-related violations are far more unexpected than motorcycle violations because how can a big and visible vehicle still had the audacity to commit violations even when knowing that they can be caught easily in the camera? Is it because they are confident that the models will fail to capture them or is it because of the weather conditions that will make the system detectors worked inaccurately?

There is a study though where their system cannot work well if the weather is bad, and if the lights are dark, making the vehicles harder to see. It was a study by Dede et al. [1] who made a mobile app system to catch six bad driving habits, like running red lights, wrong parking, and driving too close to other cars. It identified violations and logged plates, which is a reasonable proof that mobile AI-assisted enforcement holds up outside controlled setting. Their system may work properly but only for one violation type per detection event, which meant a motorcycle running a red light with no helmet and a covered plate still only generated one log entry.

Another study by Ong et al. [2] who took the fixed-camera route instead using real CCTV footage to catch red-light running and illegal overtaking. The accuracy held up reasonably and it didn't require new hardware but the approach still couldn't cover the range of violations enforcers actually deal with day to day. Because a system that is only built around one or two violation type may not catch the other ones. Like a vehicle may park illegally and cross a lane marking and then be missing a side mirror, these are three separate events.

There are also studies who focus more on classifying and identifying the road users, other than the violations they caused. One example of the study is from Ke et al. [3] who centered their study on monitoring traffic and roads. They used background modelling and SORT tracking alongside a custom trigger method to detect vehicles, and even implemented a model to classify road users into specific types like cars, trucks, and buses. However, their system's AI classifier is heavy on resource, meaning it cannot process all video frames without a trigger framework to ignore the other frames selectively. Also, their system experiences low accuracy when it fails to separate common features of the different vehicles and may group that vehicle into the same category when it is not.

The second study is from Polu [4] where his setup is centered around car-recognition using a YOLO-based object detection model. He proposed a detection system for traffic violation that is AI-powered to catch four specific road problems which are signal jumping, overspeeding, lane violations, and not authorized vehicle movements. According to the study, the tracking accuracy can drop during heavy traffic because so many vehicles may cause problems that block the camera's view. It can also struggle with poor nighttime visibility because then objects are harder to see. This means that future researchers need better low light image processing to work in the dark.

Detection Studies

When talking about detection studies in the context of traffic violation detection, most of them used YOLO Models. Diving further in reviewing them, there

are different YOLO versions that were used and for every version, they perform good or struggle depending on the version. The different YOLO versions are like newer generations of smartphones because they may all do the same main job like detecting objects but the new generation will always get more faster, smarter, and accurate.

One example of a related study from Jin et al. [9] where their team used an older YOLO version like the YOLOv4 model to detect vehicle identification and tracking of the same vehicle as they moved across multiple surveillance cameras, but found it struggled with correctly identifying or tracking vehicles when they are blocked or hidden by other cars.

On the other hand though, a study by Manoj et al. [10] proved that newer version like YOLOv8 achieves this identifying the correct vehicle by grouping these vehicles into four main categories: two-wheelers, three-wheelers, four-wheelers, and large vehicles. By grouping them, the system can separate and classify the different types of vehicles without worrying whether it is blocked or hidden by another vehicle because it could just group it into its correct category.

Another study has used an older version called YOLOv5 to detect vehicle license plate in Chinese characters in natural and complex environments [11]. However, their system has shown a weakness of failing to accurately recognize the Chinese characters from provinces other than Anhui because the training data they used was not balanced. This prove that YOLOv5 models are likely to miss very small or heavily occluded vehicles.

To contrast the weakness, a study by Aboah et al. [12] tried to catch violations with visual complexities like occlusion and perspective by using YOLOv8 and proved that it can detect accurately even under varying weather conditions.

Tracking Studies

Tracking a vehicle is a completely different aspect of difficult than just detecting a vehicle. So, that's why choosing the correct tool to use among all the tracking tools is important because it determines whether it can identify the vehicle properly and track it than just following it carelessly.

There is a study who used a ByteTrack tracking method which failed when it can only follow every detected vehicle, even if the detector is not fully confident. It was a study by Liu et al. [13] that detects the vehicle flow by tracking four classes of vehicles (cars, buses, vans, and others). They used the ByteTrack algorithm to extract the unique appearance of the vehicles but found their system failed to maintain clear and stable imagery because of the traffic environment complexities.

However, when a study by Zhao et al. [14] have used DeepSORT as their tracking method, they found that it can use the vehicle's movement and visual appearance to keep the same ID across frames, and proved that DeepSORT maintains ID during overlap. It is also noted that they built a system that extracts multi-vehicle motions which is proof that this tracking tool is far more accurate in

tracking individual vehicles than just tracking almost every detected object like ByteTrack.

When talking about tracking multi-vehicle classes, a study has achieved this by using BoT-SORT tracking algorithm [15]. Their study revolves around tracking cars, buses, trucks, and motocycles. However, their system becomes slower as more vehicles appear in the video, causing the processing speed to decrease. Although it is better at maintaining the correct ID when vehicles overlap or when the camera moves, a system that is required a stable ID tracking is more suitable with DeepSORT.

That's why for our system, a study from El-alami et al. [16] is more relevant to our study because they used DeepSORT algorithm to track and detect multiple vehicles in urban traffic. This kind of algorithm proved that it has the ability to describe what each vehicle looks like and then used that description to recognize the same vehicle again if it was lost.

A study has also proven the used of STC-SORT when detecting multi-object positions but it can only just make sure the vehicle's movement stays smooth and consistent rather than maintain the ID of the vehicle [17]. The problem with this tracking method is that it is not commonly adapted, so support and documentation are more limited.

One limitation that was noted in the existing studies is that most were conducted in traffic conditions that are different from those commonly seen on Philippine roads.

OCR (Optical Character Recognition)

Detecting a violation in a video frame is only part of what a traffic enforcement system needs to do. Without a readable plate number attached to the detected violation, there is no way to identify which vehicle was involved.

Satya et al. [18] study was one of the studies who used OCR engine to read the license plate text after improving the quality of the image of the detected vehicles. They found that it is well-suited for license plate recognition because its main purpose is to read the text from the images. Without it, a system will only know where the license plate is, not what it says.

However though, a separate study from Alvar, Uyanik, and Bajic [19] developed a system that detects the elements from a traffic scene while protecting the license plate privacy by using a method called privacy-preserving. But they noted that if the privacy setting is too strong, the system will intentionally reduce the quality of the license plate images, making the plate number difficult or impossible to read. This is why if in an instance we may use that method to enhance plate images, we must be wary of its weakness.

Research Gap and Synthesis

All the studies reviewed in this chapter have contributed for the better of a specific problem within traffic monitoring and violation detection but most have

given us the opportunity to tackle an entirely new aspect or concept for this area of study. Because although studies like Dede et al. [1], and Ong et al. [2] for car-related aspects, and Tomas and Doma [6] for motorcycle-related violations made a system that could detect individual violations only, their methodologies leave a gap because no study has attempted to build a system that can simultaneously detects multiple violations across both cars and motorcycles.

Most of the reviewed studies were built around one violation type or one vehicle class. The study closest to a combined approach detected helmet non compliance, missing side mirrors, and plate numbers together, but only for motorcycles and only from a fixed camera angle, with no continuous vehicle tracking and no coverage of car-class violations.

While studies from both detection and tracking studies proved that their models worked, they were deployed in foreign conditions or live systems, leaving a gap for offline simulation-based testing contextualized specifically for local Zamboanga traffic conditions.

All the combined studies, after reviewing, feels like it is not enough because most lacked the system of being able to detect multiple violation from both cars and motorcycles and in an offline simulation setting. This leaves us the gap to be able to build an offline prototype that can detect the following violations: no helmet, overloading, no plate number, counterflowing, no side mirrors for the primary scope of the system and for the extended scope which are failure to use signal lights, beating the red light, blocking the pedestrian lane, illegal parking, illegal u-

turn, reckless driving, and no seatbelt modelled after Zamboanga City traffic conditions.

Synthesis

Table 2: Review Matrix of Literature

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
[1]	Inefficiency of fixed traffic cameras and manual policing in detecting widespread, multi-type traffic violations	YOLOv5 + strongSORT + WPOD-NET with MobileNet	YouTube + GoPro dash cam footage	Detection accuracy, LP recognition rate	Demonstrates mobile AI enforcement feasibility	Single-camera angle; bandwidth dependent
[2]	High cost, intensive maintenance, and inefficiency of traditional traffic violation	YOLOV8 + DeepSORT + Canny Edge Detection and Hough Transform.	Real-world CCTV traffic video footage collected from online repositories	Overall accuracy of 91.28% achieved by YOLOv8x.	Automation of both lane detection and traffic light state recognition	High vulnerability to vehicle occlusion, causing incomplete lane/stop line detection.

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
	detection methods (radar guns/sensors).				without manual intervention.	
[3]	High resource constraints on edge computing devices	SORT + Multi-thread parallel system + Random Forest	Real-world traffic surveillance video footage	Achieved an average of 15.3 frames per second (FPS) on a Raspberry Pi 4 edge device	Highly efficient, decentralized multi-tasking edge intelligence	Inability to access or publish the original verification data
[4]	High urban congestion, road accidents, and fatalities worsened by traffic violations.	YOLO + Spatiotemporal anomaly detection	CCTV Footage	98.2% achieved for vehicle identification	Complete end-to-end framework	Decreased detection accuracy and frequent missed detections
[5]	Human inconsistencies associated with manual traffic rule	YOLOv8 + EasyOCR	Custom curated traffic image database	Exceptional precision for Number Plate (0.981) and solid	Novel introduction of automated rear-view mirror complian	Limited geographic data focus

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
	enforcement.			returns for No Helmet (0.883)	ce tracking	
[6]	High road accidents and fatalities in the Philippines involving motorcycle riders	YOLOv5	Custom field data	Successful classification of 791 out of 809 total detected motorcycle riders	Highly efficient YOLOv5s frame to balance execution speed	Data imbalances and sample size inconsistencies
[7]	Hard to detect two-wheeler traffic violations under low-light conditions	YOLOv8 + SMTP	Custom LoLTV dataset	Precision, Recall, Accuracy, Time per detection	First two-wheeler violation dataset tailored for low-light scenarios	Model misclassifies non-helmet headwear as helmets
[8]	High fatality rates from motorcycle accidents.	Two-stage deep learning pipeline	Custom dataset of 1,715 images.	Precision, Recall, Accuracy, mAP50	High real-time object detection precision.	High training/validation simulation times.

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
[9]	Tracking traffic loads	YOLOv4 + Kalman Filter + Hungary Algorithm	Custom Jiubao Bridge dataset	Multi-camera: >92.5% accuracy	Adaptive lane division optimizes positioning without extra math	Susceptible to occlusions/blocked cars
[10]	Vehicle detection, classification, and counting for urban traffic surveillance	YOLOv8 + OpenCV	Custom dataset of 349 total images	Real-time testing via smartphone camera webcam	High recall rate (0.92) for large vehicles	Inaccuracies when counting multiple categories of vehicles simultaneously
[11]	License plate degradation in complex, unconstrained environments	YOLOv5 + PDLPR	CCPD + PKUData	Intersection Over Union (IOU)	No manual image correction or character segmentation needed	Limited type recognition due to training data
[12]	Lack of real-time performance in	YOLOv8 + SCAN clustering	2023 NVIDIA AI City	mean Average	High precision across diverse	Computational overhead

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
	existing systems	algorithm + TTA	Challenge	Precision (mAP)	visual environments	
[13]	Traditional methods face poor small object detection	MSN-YOLOv8n + ByteTrack	UA-DETRAC subset	mAP50, MOTA, MOTP, IDF1, FP, FN, IDS, and FPS	Reduced parameters; good edge/small object handling	Struggles with environment changes like lighting and weather.
[14]	Degradation of dynamic weighing accuracy	SSD model + DeepSORT + GPS-based camera	Real traffic surveillance video from Tongchuan City, Shaanxi Province	MAE, RMSE, Accuracy and Recall	High detection accuracy (93.6%) and processing speed (66.3 FPS)	High computational resource consumption
[15]	Heavy reliance of traffic surveillance on manual observation	EfficientDet + BoT-SORT	Real-world traffic video	Vehicle Count per Frame	Stable identification of multiple overlapping	Evaluation restricted to an unspecified single video source
[16]	Inefficiency of sequential processing in two-stage object detectors	YOLOv5 + DeepSORT	Locally collected traffic video dataset	Processing Time per Frame (ms) relative to vehicle count	Exceptional scalability with near-constant frame processing times	Restricted validation on a single localized data source

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
					as traffic load spikes	
[17]	Static, context-agnostic rule-based data	STC-SORT	VisDrone + UAVDT	Evaluated on MOTA, MOTP, IDF1, IDSW, MT, ML, FP, FN, and FPS	Strong small-object and scale-variant detection	Scaling limits deployment
[18]	High computational demands of traditional ALPR systems	YOLOv8 + OCR	CCPD + UFPR-ALPR + RodoSol-ALPR	mAP@0.5, mAP@0.5:0.95, Precision, Recall, F1-score, Recognition Rate, and FPS	Enhanced OCR accuracy by up to 5.1% under low light	System struggles to generalize to uncommon license plate formats
[19]	Privacy threats from vehicle tracking	YOLOv3 + Modified information-theoretic "privacy fan"	Cityscapes dataset	mean Intersection over Union (mIoU), Root Mean Square Error	Effectively balances privacy with utility	Requires selective input resolution scaling

Study	Problem	Algorithm	Dataset	Evalation	Strength	Limitations
				(RMSE), and Character Recognition Accuracy (CRA)		

Conceptual Framework

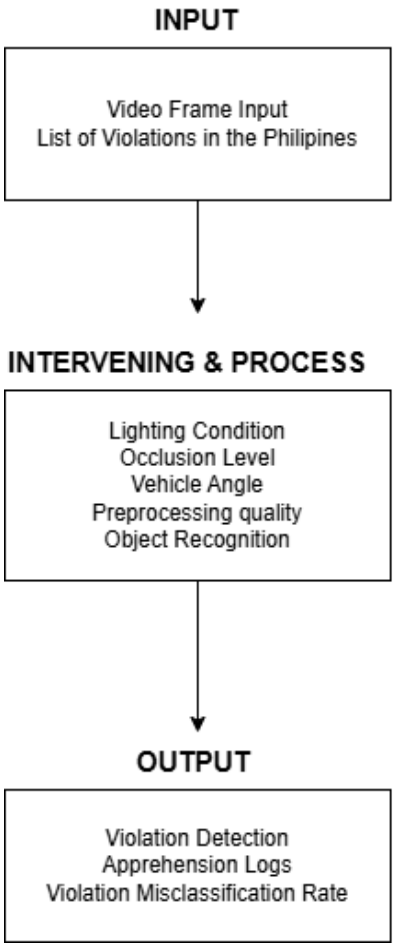


Figure 1: Conceptual Framework of the HULI-KAMOTE Automated Traffic Enforcement Platform

The conceptual framework of the HULI-KAMOTE is structured around Input-Process-Output (IPO) model.

The input element takes in raw video frames, and the list of violations in the Philippines or more specifically the top 20 violations from the LTO. The performance of the system depends in the quality of this input. If the video frames are not clear or the violation list is incomplete, the rest of the system has nothing accurate to work from.

The middle phase is where the processing is going to happen. This is where the system processes the conditions such as poor lighting, vehicles that are hidden from view by other vehicles, and plates that are captured at low angle. These are expected to affect how well the system works since these are common conditions on the Philippine roads.

The last layer is where it is intended to produce the output which are the violation record (vehicle type, violation type, and plate number where readable), the apprehension logs of violations that are captured, and the violation misclassification rate to see how often the system misclassifies a vehicle or violation.

CHAPTER III

METHODOLOGY

Research Design

This study uses Applied and Simulation-Based Empirical Research Design to develop our project HULI-KAMOTE, which is a functional and automated non-contact apprehension software platform. We are to engineer a working software system instead of relying solely on theoretical modeling in order to fulfill a concrete enforcement mandate. As for the evaluation of the platform's efficacy, we will use actual pre-recorded traffic video traces taken from our sources which is done under a controlled, offline simulated environment; This approach of simulating the process of violation detection, allows us to benchmark the system's detection accuracy and logging the fidelity against traffic violators. Furthermore, due to the offline nature of conducting our tests, we can maintain strict compliance with the Data Privacy Act (R.A.10173) by not exposing live citizens to active surveillance.

To ensure modularity of our system, we organized the development of HULI-KAMOTE into three distinct phases. This approach allows us to modify or enhance parts of the pipeline without needing to start all over again. This is due to the nature of models that might be changed in the future for alternatives or the release of new and much more reliable models, which may be used in this system for better efficacy and performance.

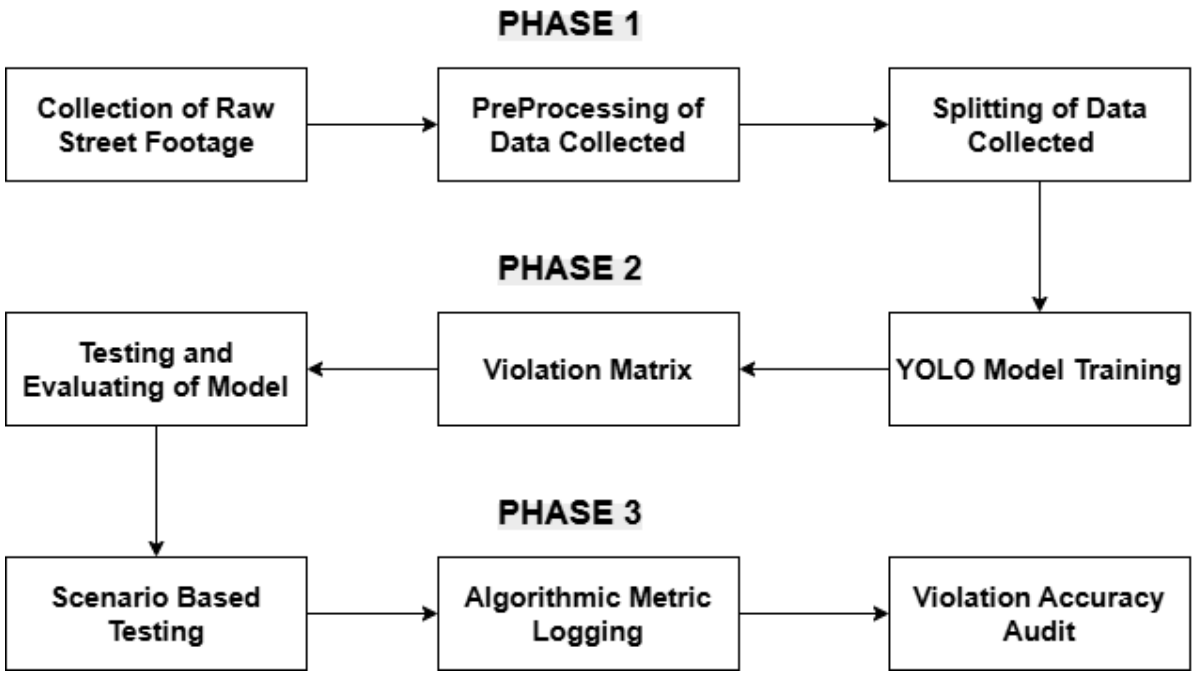


Figure 2: Operational Flowchart of the Modular Traffic Violation Detection Methodology

Phase 1: Input Preparation and Data Gathering

Initially, we focus on gathering the raw data for our system. We will prioritize institutional data from Barangay Baliwasan CCTV footages in the area, and the existing CCTV footages taken from MMDA to ensure that our models; when trained; will demonstrate effectiveness based on the Philippine road information. Since data privacy is a must, we are to comply with Republic Act 10173 by enforcing mandatory anonymization loop. We will apply Gaussian blurring to the faces of our riders and the plates of the uninvolved motorists or vehicles before

any annotation will be done, to ensure data privacy is protected even during the training stage of our chosen models.

Phase 2: System Development and Pipeline Integration

In this phase, we will build the core detection and tracking logic of our system. We are to base our software logic on our manually created Violation Matrix; which is composed of collected violations from existing traffic enforcement data in the Philippines; to ensure we can maintain consistency with the legal standards of the Zamboanga City Traffic Management Office. Modular Architecture is purposely done in each phase: Stage 1 handles the missing helmets, number of passenger for motor vehicles, along with the absence of plate number. Stage 2 uses the DeepSORT model to track the vehicle's path across the frame for violations such as illegal turning, illegal parking, counterflowing, and running on red traffic light. Stage 3 is where the extraction of license plate characters will occur upon passing the previous violation detection. With this integrated setup, it allows the system to resolve behavioral violations like counterflow while reading the violator's plate number characters.

Phase 3: Technical Audit and Expert Validation

This phase actually measures if the system is ready for road deployment. We will use the Logging and Organizational Accuracy Rate (LOAR) as our main scoreboard. This is designed to be stricty on a purpose that if a system detections violation but logs the wrong plate number, it would be worse than useless in a real

deployment and would result to false positives like ticketing oan innocent drive which may compromise the trust on our system. Additionally, we are to conduct a qualitative validation with Land Transportation Office (LTO) personnel whom of which is qualified and authorized for ticketing road violators by allowing them to interact in our dashboard and provide a verdict for its usability and effectiveness for actual deployment.

Data Collection

To train and evaluate our model, we chose to prioritize locally sourced video footages instead of generic datasets because the success of our system (HULI-KAMOTE) depends on catching violations that are unique to Zamboanga City. We aim to make our models adapt to the mixtures of vehicle found in our roads like motorcycles, jeepneys, tricycles whilst making it focus on motorcycles and car vehicles specifically. Hence, our data is being taken from three sources:

1. Barangay Baliwasan CCTV Data

Our main source is the closed-circuit television (CCTV) network from the main Office of the Punong Barangay Baliwasan by formally requesting their stored footages that accurately captures local street geometries and local traffic data flow. We are to provide our own local storage device as high quality uncompressed videos will consume a lot of space. By doing this, we aim to ensure that data acquisition from the barangay will not place a burden upon their digital infrastructure.

2. MMDA Historical Video Traces

This is our secondary source by which, we aim to acquire with our utmost capacity to acquire from the Metropolitan Manila Development Authority. Acquiring such data might take longer which could compromise our target deadline for this project, which explains why it is our second option for this study. But, acquiring such extremely high-density intersection scenes, it will train our models to be exposed with more demanding vehicle occlusion and much more higher vehicle counts than what our standard local footage from the barangay will provide.

3. Contingency Field Recording

If neither of our first two sources would every be available in worst case scenarios, we will resort to manual field recording; which serves as our insurance policy for this project and as a fallback plan. We intend to explicitly select sites in Zamboanga City that are known for high violation frequencies to increase our chances of acquiring data that we can train for our model as it requires real violation from within the footages we acquire in order to recognize violations. This should guarantee that our footages will represent an accurate perspective of a law enforcement, and it should allow us to capture "kamote" drivers safely for road violations.

Automatic Tagging Process

Upon extraction of frames from our acquired footage, we need to draw bounding boxes for every single frames of which vehicles are detected on our data which will be used to train our model. This process of manual annotation drastically slows down the development. Hence, we are to utilize an existing open-vocabulary Vision-Language Model named Grounding DINO to automatically annotate our raw footages. Grounding DINO will generate preliminary bounding boxes targeting the vehicles and license plates. This should save the researchers hundreds of hours for manual annotation.

The researchers are to be part in this process by visually verifying the accuracy of each bounding boxes. In this strict review phase, we are to assign the designated label class from our defined taxonomy. Incase of false negatives from the automated scripts, we will manually annotate the frames to ensure all necessary frames that require bounding boxes were annotated. In this exact phase, we are to set cross-frame tracking anchors which serves as a unique identifier that allows DeepSORT algorithm to trace the tracked vehicle across the sequential video frames. However, maintaining the identity track will be our main hurdle considering the dense traffic condition in Zamboanga City, hence, we will initialize a new track anchor upon a vehicle entering the frame and would retain that anchor until the vehicle leaves the camera frame or is covered by visual occlusion from a larger vehicle like trick or jeepneys.

Structured Label Taxonomy

In this study, we are to use our defined specific class taxonomy for annotation, organized by vehicle type for all training and evaluation:

Motorcycle Vehicle Classes:

• helmet_on — Rider's head region detects the presence of a helmet covering the crown, with chin strap discernible.

• helmet_off — Rider's head region has no helmet or covered only by non-protective headwear (cap, bandana, or similar).

• rider_bounding_box — Bounding box enclosing the primary rider seated at the front of the motorcycle.

• pillion_bounding_box — Bounding box enclosing any additional passenger seated behind the primary rider. Multiple pillion_bounding_box instances on a single motorcycle frame indicate an excess passengers or overloading violation (more than one pillion passenger is present beyond the vehicle's licensed two-occupant capacity).

• license_plate_zone — Bounding box localized to the motorcycle's rear or front-mounted license plate region, used as the input crop for the automated license plate character recognition engine.

Car Vehicle Classes:

• car_bounding_box — Bounding box enclosing the full vehicle body of a detected private automobile, defining the spatial region within which all carspecific sub-region classifiers operate.

• license_plate_zone — Bounding box localized to the car's rear or front license plate mounting position, used as the input crop for the automated license plate character recognition engine.

Traffic Violation Identification and Technical Detection Matrix

Below is the table inventory that this study plan to target, depending on the data we may collect. It is organized by its label classes, followed by the sample images to visualize the violation and the challenges to build it.

Table 3; Traffic Violation Detection Matrix

Classes to be Detected & Legal Basis	Sample Images	Implementation / Computer Vision Challenges
MOTORCYCLE — No Helmet / R.A. 10054 (Motorcycle Helmet Act of 1993): helmet_off, rider_bounding_box		Low resolution makes it hard to see where a helmet ends. Our model often confuses dark bandanas or backward caps with actual helmets, which creates detection errors. If a passenger's arm or a large delivery box blocks the rider's head, the system will lose the target zone entirely.

MOTORCYCLE — No Side Mirror / R.A. 4136 (Land Transportation and Traffic Code), Sec. 34: mirror_absent, rider_bounding_box		Side mirrors are small to see in the camera that's why there are difficulties when trying to detect if a mirror is missing. What more is that objects near the handlebars, such as brake levers and colorful decorations can be mistaken for mirrors.
MOTORCYCLE — Excess Passengers / Overloading / R.A. 4136, Sec. 54(f): rider_bounding_box, pillion_bounding_box (count > 1)		Excess passengers mean that there is an additional person who is sitting in the motorcycle and since when riders are sitting close together, the system may mistake it as one person only. The same as large sacks of rice or heavy bags carried on motorcycles because motorcycles in Zamboanga often carry those. It is also difficult to separate multiple riders when the motorcycle is moving fast because the image becomes blurry.

MOTORCYCLE — Counterflow / Driving Against the Flow / R.A. 4136, Sec. 45(b): Spatial-temporal tracking trajectory across annotated lane direction zones		When counterflowing, the riders are often blocked by a truck or a large vehicle from the camera's view, making the system lose track of it and fail to detect the violation
MOTORCYCLE — Illegal Turns / R.A. 4136, Sec. 42(d); MMDA Regulation No. 96-009: Spatial-temporal trajectory analysis across annotated turn-restriction zones		Footages from older barangays are often low in quality, making turn detection more difficult. This could also make the system have difficulties telling the difference between an illegal turn and a rider who is only changing a position to avoid a pothole.

MOTORCYCLE — Illegal Parking / R.A. 4136, Sec. 51; Local Ordinance No. SP-1160: Stationary vehicle persistence within annotated no-parking zones		The system may mistake a motorcycle dropping off a passenger as parked motorcycles in a single frame. Frequent short stops may also cause the system to detect the parking violation incorrectly.
MOTORCYCLE — Disregarding Traffic Signs / R.A. 4136, Sec. 42(a): Trajectory crossing a stop-line or signal-state zone during a restricted signal phase		Riders often approach a stop-line and decelerate without actually crossing. These "creeping" behaviors at Zamboanga intersections produce proximity-based false positives where the system thinks a violation is about to happen.
MOTORCYCLE — No Plate Number / R.A. 11235 (Motorcycle Crime Prevention Act); R.A. 8750 (Seat Belts Use Act of 1999 analog for plates): license_plate_zone returns null character output from the automated license plate character recognition engine		Plates that are covered in mud after a rainy weather or hidden under a mudguard may make it hard to see in the camera. It is also small and difficult to read that the system may produce uncertain results that need to be checked manually.

CAR — Illegal Turns / R.A. 4136, Sec. 42(d); MMDA Regulation No. 96-009: Spatiotemporal trajectory arc across annotated turn-restriction ROI		The system may mistake the vehicle that was only changing lanes or making a legal wide turn as illegal turning when it is not. This may cause the system to detect a violation when there is none.
CAR — Illegal Parking / R.A. 4136, Sec. 51; Local Ordinance No. SP-1160: Stationary persistence within no-parking ROI		Double-parked vehicles often block each other so the system may find it hard to identify which one is violating the no-parking rule.
CAR — Disregarding Traffic Signs / R.A. 4136, Sec. 42(a): Centroid crossing stop-line during restricted signal phase		Large vehicles may slowly move past the stop line even when the driver is trying to stop properly. To prevent the large vehicles such as trucks and vans from being incorrectly detected as violating the stop line rule, the system will track the vehicle's center instead of its front edge.

CAR — Driving Against the Flow / R.A. 4136, Sec. 45(b): Spatiotemporal directional trajectory against annotated lane-flow reference		In busy roads, drivers may back up to make space which can confuse the system because it can look like they are driving the wrong way when they are not.
CAR — No Plate Number / R.A. 8750; R.A. 4136, Sec. 18: license_plate_zone returns null output or zero-confidence character extraction		Rainy season is always the challenge when detecting license plates because most of it will get covered in mud and dirt, making it harder to capture and read. In cases like these, the system must be able to distinguish between a vehicle with no license plate and a vehicle with an unreadable plate because these situations require different violation records.

Algorithmic Framework

The pipeline of HULI-KAMOTE is structured as a three stage computational framework where the output of each phase feeds directly into the next phase. Stage One will be the scanning of each video frames to detect vehicles and identify violations. Stage Two will monitor continues spatial movement of the vehicles over a specific duration of time. Stage Three will be activated upon confirmation of

violation to extract the alphanumeric license plate characters of the vehicle. With this structure, we can isolate algorithmic errors within individual pipeline before reaching and corrupting the final administrative log.

Stage 1: Spatial Object Detection and Attribute Localization

Our main model will aim to detect all target objects within a single isolated video frames for every single frames we have. The AI will track the position of every single vehicle within the frame and their positions while simultaneously scanning for target objects like helmets and plate number for motorcycles. Our system's Boolean evaluation logic is simple enough that it flags whenever the bounding boxes fail to detect the target object that must be present like helmets. And in order to detect the number of passengers, we are to use the same mechanism of counting the number of heads for detection of helmets and allows our model to flag for overloading of passengers. The plate detection is also being implemented within the bounding box of the vehicle, and once absent, it will automatically flag the "No Plate Number" violation. Lastly, to avoid messy data being sent to the next stage, this stage uses Non-Maximum Suppression (NMS) filter to remove duplicate or overlapping detections that are targeting the same vehicle and guarantees mathematically clean and verified frame records for our temporal tracking algorithm.

Stage 2: Spatiotemporal Multi-Object Tracking

Relying on a single frame cannot verify whether the vehicle is on an illegal parking violation, or counterflowing and alike. Hence, we are to utilize DeepSORT algorithm to do the task. What it does is it tracks our target by fusing the spatial positioning of each vehicle along and remembering their exact appearance even after the model losing sight of the vehicle within a limited timeframe. This allows the system to track the movement of the and effectively flagging for any violation that requires movement tracking for cases of counter flow, illegal parking, and illegal turning. To prevent false positives, the system only flags the violation if the vehicle's wrong driving direction is sustained after a certain defined period, hence, ignoring the quick swerves in cases of avoiding potholes or road obstacles. This stage also tracks the vehicles path upon crossing the stop-line during red-light timer and flags it as red-light violation.

Stage 3: Automated License Plate Recognition and Alphanumeric Text Extraction

In this stage, the system will be translating the detected license plate zone into a much more machine-readable alphanumeric strings for vehicle identification logging. The cropped frame will first undergo under image preprocessing before optimal character recognition (OCR) to prevent unwanted results from poor quality frame resulting from severe motion blur or heavy video compression artifacts. This two-step process allows the system to effectively extract texts much more accurately by localizing first the plate number area, preventing any background noise which might in turn affect the accuracy of plate number alphanumeric character detection such as existing objects that contains texts like shop logos,

vehicle texts, and pedestrian signage. Lastly, upon successful extraction of text, the system will then merge the plate number identification with validated violation class and the results from persistent DeepSORT track ID, therefore, producing the complete legal non-contact apprehension of road violators which are logged as an evidence record in our administrative dashboard.

System Architecture And Dashboard Design

For this thesis project, a dashboard will be provided designed to help simplify the complex detection pipeline into a much more friendlier and easy to operate approach for our human operators. It will serve as the heart of our entire system where all of the processes will occur along with the necessary human intervention. The dashboard is capable of translating mathematically raw algorithmic outputs into something much more readable and verifiable evidences for ease of operation. The Traffic Officers which would validate and use this project should be able to easily understand and leverage these outputs upon actual deployment of the system, which leads us to building the system's user interface into three distinct functional areas: One will handle real-time monitoring, another will manage the logging of violation which is done automatically by the system, and the final area allows the operators to have their administrative review capabilities. A simplified version of the general data statistics and analytics page will also be provided for operators to have their insights regarding the information being outputted by the system and the statistical numbers of the road violations occurring.

Violation Detection Subsystem

This subsystem will give our operators the access to observe the software working in real time. It will process incoming video feeds from our raw footage taken from our specified sources and will immediately overlay visual metadata on the live screen for our targets. Every bounding boxes will be under visual hierarchy which are strictly color coded based on their violation severity for ease of distinct visual identification and distinguishing of violations by our operators. The display in the UI will also feature integrated label banners that explicitly shows the specific violation class along with the model's mathematical confidence score upon detection. A live status bar is also to be implemented to track the Frames Per Second (FPS) along with the active vehicle counts in each of our video frames. All of these is to aim for a smooth monitoring of our system's processes for any unwanted results, system lags, and unexpected issues that may hinder the effectiveness and efficiency of our system.

Vehicle Identification And Reecording Subsystem

Upon system detecting and confirms the violation for the physical attributes needed violation or the movement tracker violation, this would automatically be written in a database as a record, aiming for data integrity as our absolute priority during this process. This subsystem will follow a defined strict sequence to construct a valid legal log entry for our enforcers to validate. Firstly, the specific violation class and the exact millisecond timestamp will be noted down. Second, the alphanumeric plate number string extracted from our optical recognition engine

will be pulled for the records. In cases of environmental factors affecting the OCR engine to fail to identify the plate number, the system will mark it as "NO PLATE" into our database instead of a blank column box which may be verified by the enforcers. Although it may look like a software failure for our enforcers, it also, on the other hand, confirms explicitly that our system successfully did the optical check with our OCR model but hasn't found any readable texts or characters from the extracted cropped frames. Lastly, this subsystem will also map the vehicle's persistent track ID directly to the database record to ensure the logged violation connects precisely to the correct vehicle trajectory on the frame recorded.

Results Review Subsystem

This is the final area of our subsystems that would serve as a searchable archive feature for our project where the operators of the system can view every violation logged during the session run. A sorting and filtering features will be included for efficient look ups of records instead of scrolling through a long list of logged violations. This is feature or subsystem is also vital for identifying repeating "kamote" violators that habitually violates the Zamboanga's traffic laws. The operators can view the specific evidence frame; the specific moment the violation was flagged along with its tied metadata; and verify the record an apprehension notice will be issued to the violators. To abide and stay compliant with the Data Privacy Act (RA10173), we will implement the PII masking as a security protocol to prevent any raw, unasked footage to be extracted from the database; ensuring

only the masked visual evidences and plain text plate number characters are accessible to the authorized personnel.

Evaluation Metrics and Performance Benchmarking

The evaluation of HULI-KAMOTE will be based on our core question in this research: "How accurately does our system log and organize the violation in detects?" This is beyond with just checking if the detection models draw the correct boxes, identify the incorrect vehicle trajectory, extracting the plate number, rather, we measure the entire process by verifying if the system successfully converts our target video frames into an accurate database record. All records logged in the system must be complete and accurately linked for the data to be valuable.

With that, this study uses Precision and Recall to come up with an evaluation result on how well our model identifies road violations:

Precision helps us detect if our system is raising false positives. The higher the precision leads to us knowing that the flag is reliable and legally credible as false alarms or false positives results (an instance where the system flagged a violation but no violation/s has occurred) tend to result in a more burden manual review for the enforcers.

Recall helps us know if our model missed true violations (false negatives — an instance where the mode incorrectly flagged as no violation but violation does exist on that particular vehicle). A low recall score is an enforcement gap for

this study where the camera captured the violation but the model missed the detection.

Having these two parameters, we aggregate them into mean Average Precision (mAP@50)

As for our OCR character recognition, we will use **Character Error Rate (CER)** and **Word Error Rate (WER)** to test the accuracy of character extraction for our plate reading feature

Character Error Rate (CER) simply measures the accuracy at the character level whilst Word Error Rate (WER) treats the entire extracted plate string as one unit. WER is a crucial metric in this study as if a single character goes wrong, the identification might be pointed out to the wrong vehicle plate number which will result in ticketing innocent drivers.

Logging and Organizational Accuracy Rate (LOAR)

LOAR is the primary system-level benchmark of this study as it measures the proportion of violation events that happens to have been correctly recorded in our database from our total number of actual violations that the AI caught in the video. LOAR is designed to be strict in purpose by only counting the correct the records if and only if the violation type, the track ID, and the plate number character string all match at the same time. Each of the components may work as fine as they are independent based on what stage they are in, however, LOAR exposes

the final call regarding the existence of integration failures. This metric ensures that only the records with quality of evidence can pass within the system detection.

Testing and Evaluation

We will test the HULI-KAMOTE using two distinct layers: We will perform first a Technical Simulation Benchmarking by running our system against an evaluation set of annotated video frame footage we have on hand where it could provide a fair and reliable measure of the system's accuracy in detecting traffic violations under different environmental conditions. In this layer, the LOAR is only as reliable as our own manually annotated collection of frames.

The next layer is where the actual enforcements come in play which is the Third-Party Expert Validation. We will be working with the Land Transportation Office (LTO) personnel in this phase. They will be watching real-time detections happening from our system's dashboard and examine the complete records afterward. Using their authority, they are to come up with a verdict if our software is ready for the road deployment. Our evaluators will be completing a standardized form that generally targets the parameters such as the FURPS which stands for function Functionality, Usability, Reliability, Performance and Supportability. With their professional assistance, we can verify if our logs would accurately support a non-contact apprehension of road violations under the current LTO workflows.

Evaluation Criteria for Expert Validation

Criterion A - Functionality and Detection Accuracy: This criterion shall cover our detection for missing helmets, overloading passengers, missing plate numbers and alongside with other violations movement-based violations. The LTO officers will help us confirm if our software accurately isolated these evens as accurate as how a human enforcer would in actual road conditions using their authorities.

Criterion B - Reliability and Performance: This criterion will be added to measure our system's processing speed even under heavy occlusion instances where traffic gets heavy. We also would ask our experts to check if our error rates are acceptably low in preparation for deployment.

Criterion C - Usability of the Dashboard: In this criterion, we check if the user interface is simple and easy enough to interpret for the enforcers even without technical background. We will also evaluate in this criterion whether or not the evidence frames and the timestamp provided are sufficient information for documentation purposes without having the need for manual audit.

Criterion D - Evidentiary and Enforcement Value: Our evaluators will be using this section to decide if our captured video frames and plate number string characters provides proof that is required for non-contact apprehension. We will also ask them in this category to verify our logging methodology if it is compatible with the current LTO enforcement workflow

Criterion E - Data Privacy and Security: This criterion will evaluate our Data Privacy protocols and our efforts to handle Personally identifiable Information (PII) correctly. We will show to our evaluators regarding the automated face blurring and plate number masking processes along with the strict guard of mandatory censoring of footages before extraction.

Criterion F - Overall Acceptability: the final section of our evaluation will be the generally verdict of the system's potential. Our main goal is to verify if our HULI-KAMOTE is truly ready for real-time deployment. We will ask our evaluators to weigh everything they encountered within our system—along with the technical limitations—to come up with a conclusion with their expertise whether or not this tool is viable and usable for future automated non-contact apprehension of traffic violation enforcement Zamboanga City

Statistical Analysis and Analytical Approach

To process scores from our LTO evaluation forms gathered from our evaluators, we will be utilizing a structured statistical approach. This ensures that the Likert Scale ratings for instance that are given to the experts will be translated into clear and actionable conclusions for each criterion. This approach helps us make our subjective feedback forms as objective as possible through aggregating these scores and ensure that we can mathematically prove whether HULI-KAMOTE is viable for standards required for actual deployment.

Weighted Mean Computation

Every single item and criterion-level scores will be using the Weighted Mean, which mainly accounts for how each of the individual ratings are distributed across all response categories instead of treating them as equal contribution in a flat average

Descriptive Interpretation Intervals

Table 4: Range of Mean Scores and Qualitative Interpretations for the Four-Point Likert Scale

Weighted Mean Range	Qualitative Descriptor	Operational Interpretation
3.50 – 4.00	Highly Acceptable	Fully meets enforcement and operational standards in this criterion, no modification required. Suitable for deployment consideration or a live piloting phase.
2.50 - 3.49	Acceptable with Minor Revisions	Meets the core standard but has specific, addressable gaps that should be fixed before formal deployment. Revisions are incremental.
1.50 - 2.49	Acceptable with Major Revisions	Shows sufficient foundational capability but needs substantial redesign of one or more components before it's operationally suitable. Cannot pass without significant further development.
1.00 - 1.49	Not Acceptable	Does not meet the minimum operational standard in its current form. Needs a fundamental rethink of the approach for this criterion before re-evaluation.

These intervals will be applied uniformly throughout every criterion along with the acceptability score from Criterion F. In cases where any of the criterion scoring below 3.50, this becomes an action item for the researchers for the post-

evaluation development cycle, taking account of the evaluations' feedback which serves as the primary guideline for our targeted improvement.

Ethical Considerations and Data Anonymization

Having the real-world video CCTV footage from Zamboanga City means the researchers are responsible for the data privacy of every person and vehicle on each of the frames we acquired from raw footages. Republic Act 10173 is the ground basis for our data policy, and due to the fact that license plates and faces are Personally Identifiable Information (PII), the system will be employing strict protocols within its workflow from the very initial stages.

Institutional Data Request and Consent

Acquiring CCTV footage from the Office of the Punong Barangay of Baliwasan will only happen after the formal signing agreements of data-sharing process. These agreements shall act as our legal binding contract which specifies which exactly data we will be utilizing for our system, training our models, and testing, who from among our team will have access to the encrypted storage, and how are we going to handle the discarding and destroying of the footage within thirty days of our final evaluation. This request letter is treated as an opportunity to build local trust with the Barangay by explaining our data privacy protocols in the early stages all set up and ready for the entire duration of the study.

Mandatory Anonymization Loop

HULI-KAMOTE uses a mandatory PII masking feature that serves as the safety protocol of guard-wall for our database. There shall be no raw, unmasked video frame that ever enters our persistent archive by ensuring that every face detected in the footage and every license plate for non violators will have Gaussian Blurring applied upon them and have their PII protected. The only downside here would be the human error, hence, we must secure thoroughly that our scripts run effectively every single session to prevent unwanted results of failing to blur faces and saved to the database as a mistake. This protocol also ensures that the people outside of this study which are included like the by standers will have their PII also protected.

CHAPTER IV

RESULTS AND DISCUSSION

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Conclusion

Recommendations

References

[1] D. Dede, M. A. Sarsıl, O. Altıntaş, A. Shaker, and O. Ergen, "Next-gen traffic surveillance: AI-assisted mobile traffic violation detection system," arXiv preprint arXiv:2311.16179v1, Nov. 2023.

[2] C. S. Ong, T. Connie, and M. K. O. Goh, "Traffic Violation Detection Using Computer Vision Techniques," *JOIV: Int. J. Inform. Visualization*, vol. 8, no. 3-2, pp. 1515-1527, Nov. 2024.

[3] R. Ke, C. Liu, H. Yang, W. Sun, and Y. Wang, "Real-Time Traffic and Road Surveillance With Parallel Edge Intelligence," *IEEE J. Radio Freq. Identif.*, vol. 6, pp. 693-696, 2022.

[4] O. R. Polu, "AI-powered traffic violation detection using CCTV footage," *International Journal of Science and Research (IJSR)*, vol. 13, no. 3, pp. 1956–1961, Mar. 2024, doi: 10.21275/SR24038113847.

[5] N. V. HO, A. Agarwal, and M. Moharir, "A Novel AI-Driven System for Real-Time Detection of Mirror Absence, Helmet Non-Compliance, and License Plates Using YOLOv8 and OCR," in *Proc. 12th Int. Conf. Emerging Trends Eng. Technol. Signal Inf. Process. (ICETET SIP)*, Nov. 2025, pp. 1-6, arXiv:2511.12206v1 [cs.CV].

[6] J. P. Q. Tomas and B. T. Doma, "Motorcycle Helmet Detection and Usage Classification in the Philippines using YOLOv5 Algorithm," in *Proc. 5th Int. Conf. Comput. Intell. Intell. Syst. (CIIS)*, Nov. 2022, pp. 21–25, doi: 10.1145/3581792.3581796.

[7] S. Bose, M. H. Kolekar, S. Nawale, and D. Khut, "LoLTV: A Low Light Two-Wheeler Violation Dataset With Anomaly Detection Technique," *IEEE Access*, vol. 11, pp. 124951-124961, 2023, doi: 10.1109/ACCESS.2023.3329737.

[8] U. U. Deshpande *et al.*, "Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8," *Front. Artif. Intell.*, vol. 8, art. no. 1582257, Jul. 2025, doi: 10.3389/frai.2025.1582257.

[9] T. Jin, X. Ye, Z. Li, and Z. Huo, "Identification and Tracking of Vehicles between Multiple Cameras on Bridges Using a YOLOv4 and OSNet-Based Method," *Sensors*, vol. 23, no. 12, p. 5510, Jun. 2023, doi: 10.3390/s23125510.

[10] M. Manoj, M. A. S, M. N. Y, S. K, Mohana, and R. K. P, "Vehicle Detection, Counting and Classification using YOLOv8 Deep Learning Model and OpenCV for Urban Traffic Management System," in *Proceedings of the 9th International Conference on Communication and Electronics Systems (ICCES)*, Bengaluru, India, 2024, pp. 2085-2090, doi: 10.1109/ICCES63552.2024.10860266.

[11] L. Tao, S. Hong, Y. Lin, Y. Chen, P. He, and Z. Tie, "A Real-Time License Plate Detection and Recognition Model in Unconstrained Scenarios," *Sensors*, vol. 24, no. 9, p. 2791, Apr. 2024. DOI: 10.3390/s24092791.

[12] A. Aboah, B. Wang, Y. Adu-Gyamfi, and U. Bagci, "Real-time Multi-Class Helmet Violation Detection Using Few-Shot Data Sampling Technique and YOLOv8," in *arXiv preprint arXiv:2304.08256*, Apr. 2023.

[13] J. Liu, Y. Xie, Y. Zhang, and H. Li, "Vehicle Flow Detection and Tracking Based on an Improved YOLOv8n and ByteTrack Framework," *World Electric Vehicle Journal*, vol. 16, no. 1, p. 13, Dec. 2024, doi: 10.3390/wevj16010013.

[14] Y. Zhao, C. Ren, S. Zhao, J. Yao, X. Li, and M. Wang, "Study of Multi-Objective Tracking Method to Extract Multi-Vehicle Motion Tracking State in Dynamic Weighing Region," *Sensors*, vol. 25, no. 10, p. 3105, May 2025. doi: 10.3390/s25103105.

[15] P. Sodegaonkar and R. A. Khan, "Vehicle Detection and Tracking using EfficientDet and BoT-SORT," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 14, no. 4, pp. 4846-4854, Apr. 2026. Available: https://doi.org/10.22214/ijraset.2026.80273.

[16] A. El-alami, Y. Nadir, and K. Mansouri, "A Modified Lightweight DeepSORT Variant for Vehicle Tracking," *(IJACSA) International Journal of Advanced Computer Science and Applications*, vol. 15, no. 10, pp. 658-664, 2024.

[17] Z. Ma, C. Chen, J. Chen, and Y. Jiang, "STC-SORT: A Dynamic Spatio-Temporal Consistency Framework for Multi-Object Tracking in UAV Videos," *Applied Sciences*, vol. 16, no. 2, p. 1062, Jan. 2026, doi: 10.3390/app16021062.

[18] B. Satya, D. Manongga, Hendry, and A. Aminuddin, "Optimized YOLOv8 for Automatic License Plate Recognition on Resource Constrained Devices," *Engineering, Technology & Applied Science Research*, vol. 15, no. 2, pp. 21976–21981, Apr. 2025, doi: 10.48084/etasr.9983.

[19] S. Ranjbar Alvar, K. Uyanik, and I. V. Bajić, "License Plate Privacy in Collaborative Visual Analysis of Traffic Scenes," arXiv preprint arXiv:2205.01724v1, May 2022.

Appendix A: Data Collected

The letter below is being prepared for submission to the Office of the Punong Barangay of Barangay Baliwasan, Zamboanga City, to formally request traffic footage for use in this study. The request is part of the data collection process, which was ongoing at the time of writing. The researchers are also pursuing the MMDA request through the eFOI Portal and CARLA-generated footage as additional sources in case institutional footage becomes unavailable.

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DEPARTMENT OF COMPUTER SCIENCE
Zamboanga City

June 19, 2026

HON. MA. JAMIELY CZARINA N. CABATO
Punong Barangay / Barangay Captain
MR. MICHAEL GREGORIO
Barangay Secretary
Office of the Punong Barangay
Barangay Hall, Barangay Baliwasan
Zamboanga City

SUBJECT: FORMAL REQUEST FOR ACCESS TO CCTV FOOTAGE — ACADEMIC RESEARCH

Dear Madam/Sir:

We, the undersigned undergraduate researchers from Western Mindanao State University, respectfully submit this letter in connection with our ongoing thesis project. Under the guidance of our thesis adviser, we are reaching out directly to Barangay Baliwasan, as your closed-circuit television (CCTV) infrastructure holds the specific footage necessary to complete our study.

Our thesis is formally titled:

"HULI-KAMOTE: Camera-Based Multi-Violation Detection and Non-Contact Apprehension System for Urban Traffic Enforcement in Zamboanga City"

The study aims to develop a computer vision-based traffic violation detection system capable of identifying common road violations — including counterflow driving, illegal turning, helmet non-compliance, overloading, and missing side mirrors — using artificial intelligence technologies such as YOLOv8 object detection, DeepSORT multi-object tracking, and EasyOCR for license plate recognition. The system is designed to operate in a simulation-based environment and is intended solely for academic demonstration and evaluation.

In this regard, we humbly request permission **to obtain pre-recorded CCTV footage** from cameras maintained by Barangay Baliwasan. Specifically, we are requesting access to available footage **regardless of specific dates of recording**, as our primary objective is not to document particular incidents, but rather to capture representative, non-incident-specific structural traffic flows that will serve as the basis for our simulation calibration and model validation. We fully acknowledge that such video records are expected to be significantly large in file size, and we wish to assure your office that our research team has already prepared the necessary high-capacity storage infrastructure to accommodate large data volumes, ensuring that no strain or burden will be placed upon the barangay's own data systems or storage resources.

The footage shall be used exclusively for the following academic purposes:

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College of Computing Studies
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Zamboanga City

1. Calibrating and validating simulation parameters (e.g., lane configurations, vehicle spacing, and traffic density) against real local road conditions in Zamboanga City;

2. Testing the detection accuracy and robustness of our trained model against actual Philippine roadside CCTV conditions, including factors such as camera angle, lighting variation, and image compression artifacts; and

3. Generating annotated datasets for academic benchmarking within a fully offline and controlled research environment.

We give our solemn assurance that the requested footage shall be used strictly and exclusively for the academic research purposes stated herein. The footage will not be used for any commercial purpose, live enforcement, public disclosure, or dissemination in any form without the express written authorization of the barangay. Furthermore, all footage will be processed through an automated privacy protection protocol — including facial blurring and license plate masking — prior to any annotation or algorithmic processing, in full compliance with Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012.

Moreover, we wish to emphasize that our research team is fully prepared to provide the necessary manpower to assist with all technical work involved in the retrieval and transfer of data. We are entirely willing to personally undertake the downloading, filtering, and sorting of the relevant video clips at a time and arrangement most convenient to your office and staff. Our team will manage the complete manual transfer of the footage using our own external high-capacity storage drives, thereby ensuring that no operational inconvenience, workload, or strain will be imposed upon the personnel or digital infrastructure of the barangay at any stage of the process.

We likewise express our full willingness to comply with any requirements, policies, or procedures that the barangay may impose in connection with this request. Should your office require additional documentation — such as a university endorsement letter, thesis proposal, researcher identification cards, or a Data Privacy Agreement — we are prepared to submit the same at your earliest convenience.

We sincerely hope that this request merits your kind and favorable consideration. Your assistance will be invaluable to the successful completion of our thesis and will contribute meaningfully to the academic community's ongoing efforts to develop data-driven solutions for urban traffic safety. Should you require any further information or clarification, please do not hesitate to reach us through the contact details provided below.

Thank you very much for your time and assistance.



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Respectfully submitted,

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Emman Nicholas Blabe B. Idulsa
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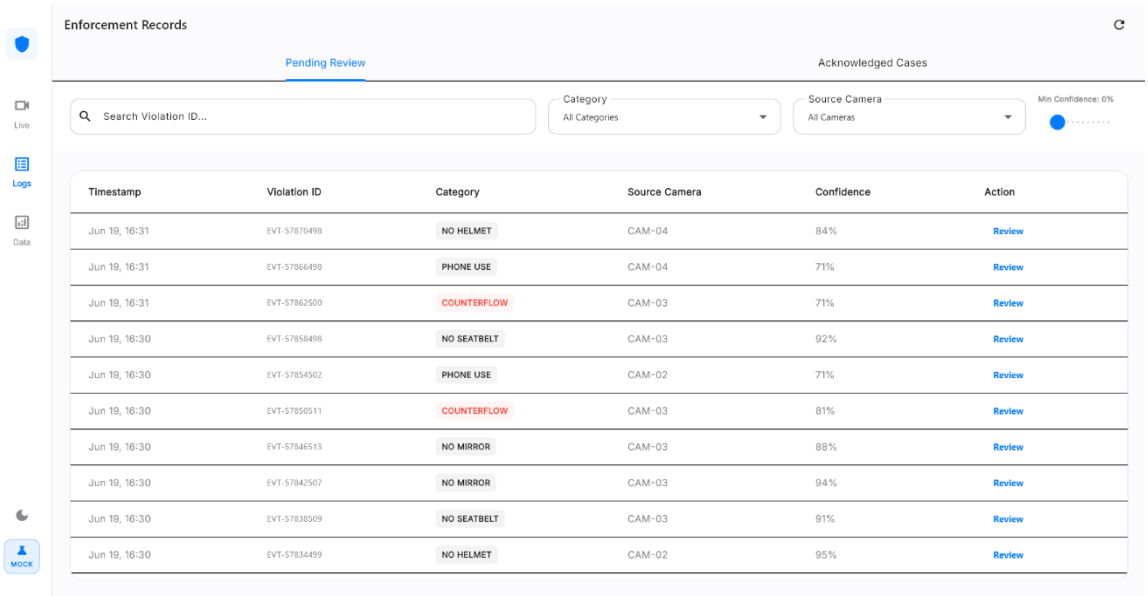
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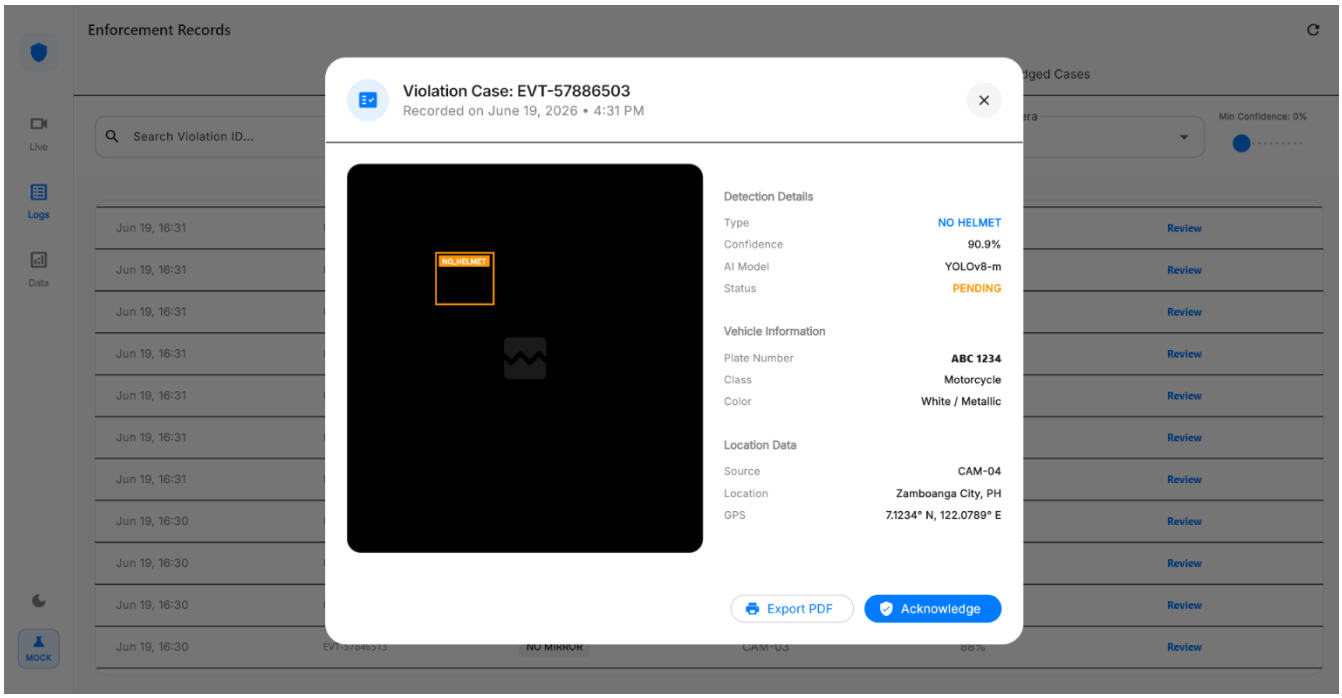
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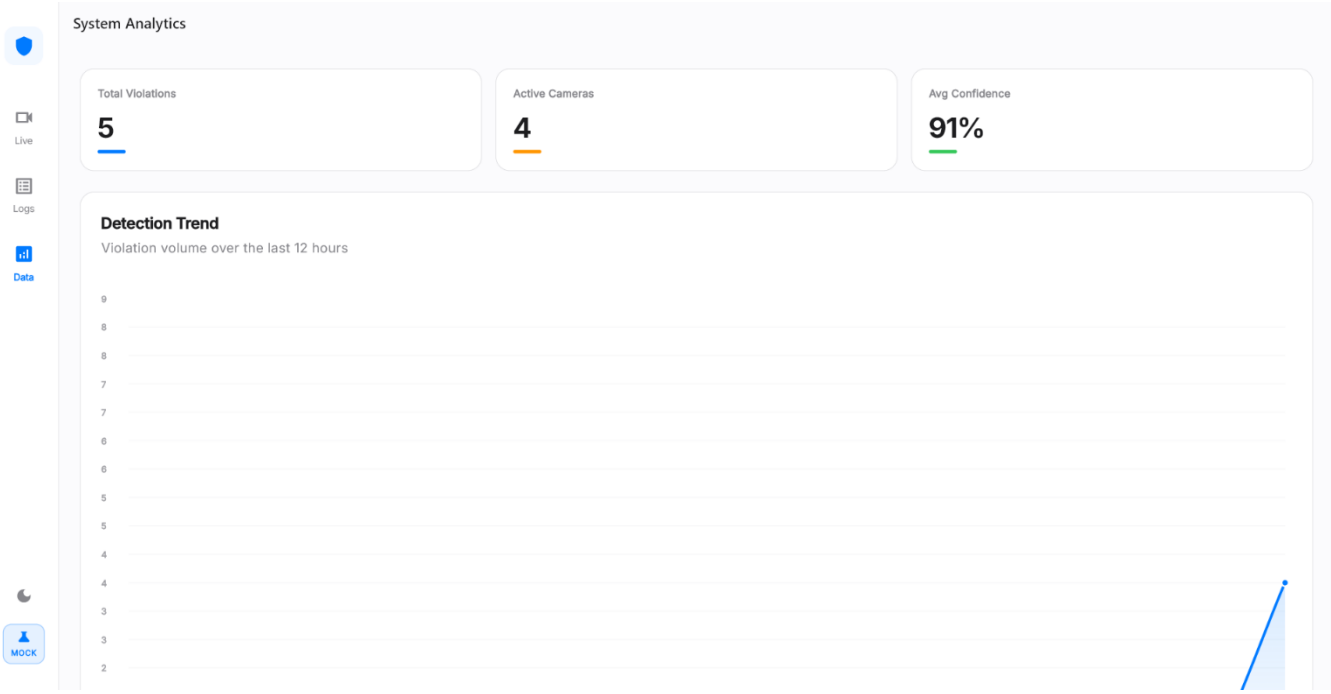
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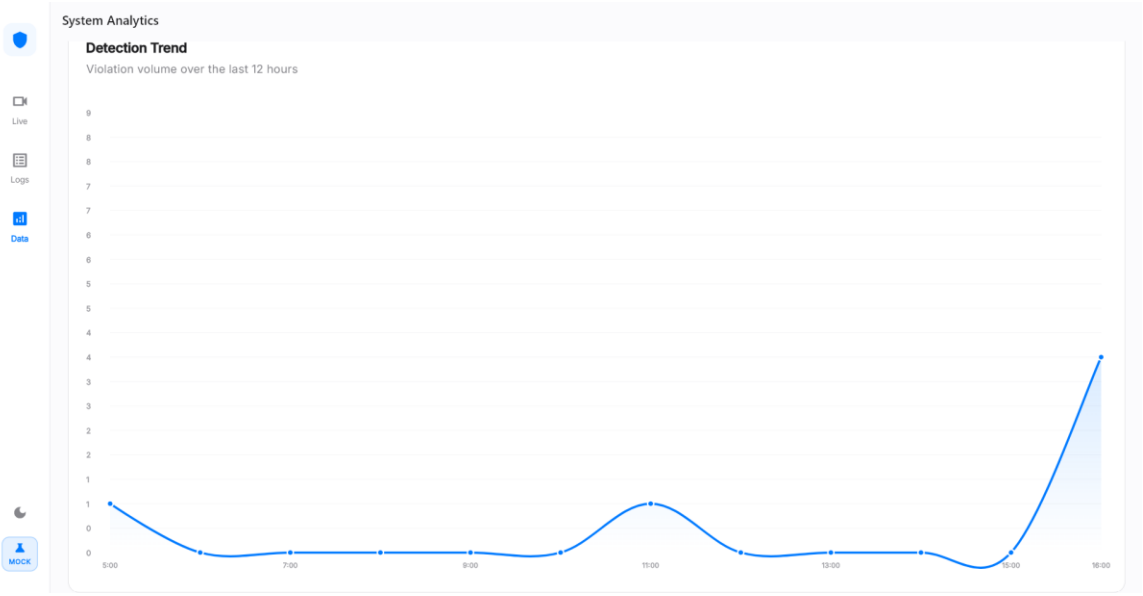
Appendix B: System Design

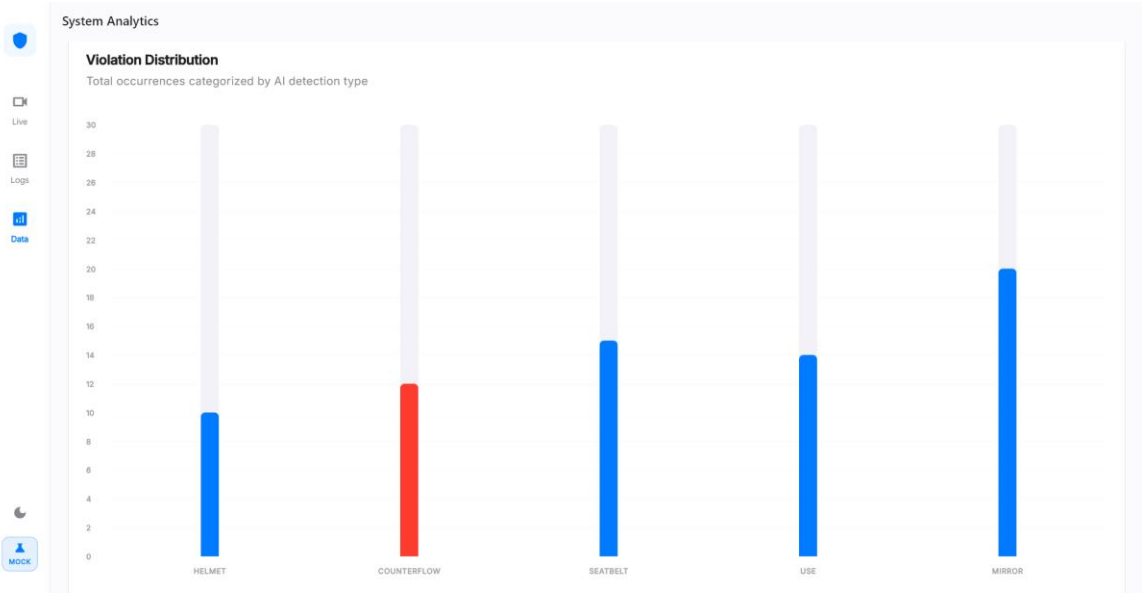





Enforcement Records
Violation Case: EVT-57886503
Recorded on June 19, 2026 • 4:31 PM
NO_HELMET
Detection Details
Type NO HELMET
Confidence 90.9%
AI Model YOLOv8-m
Status PENDING
Vehicle Information
Plate Number ABC 1234
Class Motorcycle
Color White / Metallic
Location Data
Source CAM-04
Location Zamboanga City, PH
GPS 7.1234° N, 122.0789° E
Export PDF
Acknowledge
Search Violation ID...
Min Confidence: 0%
Review
Live
Logs
Data
MOCK


System Analytics
Total Violations
5
Active Cameras
4
Avg Confidence
91%
Detection Trend
Violation volume over the last 12 hours
Live
Logs
Data
MOCK


System Analytics
Detection Trend
Violation volume over the last 12 hours
Live
Logs
Data
MOCK
5:00
7:00
9:00
11:00
13:00
15:00
16:00


System Analytics
Violation Distribution
Total occurrences categorized by AI detection type
Live
Logs
Data
MOCK
HELMET
COUNTERFLOW
SEATBELT
USE
MIRROR

Appendix C: Evaluation Tool

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SYSTEM EVALUATION FORM

FURPS Criteria and System Usability Scale (SUS)

(For Land Transportation Office Evaluators)

Date: _____

Research Title:	HULI-KAMOTE: Camera-Based Multi-Violation Detection and Non-Contact Apprehension System for Urban Traffic Enforcement in Zamboanga City
Proponents:	Emman Nicholas Blabe B. Idulsa, Jehana H. Kairan, Fatima Sheena Sharifa T. Abubakar
Main Objective:	
Venue/Site of Testing:	

Evaluator's Profile

Name:	
Position/Designation:	
Office/Station:	
Years in Service:	

This form is intended to gather the assessment of Land Transportation Office (LTO) personnel regarding the HULI-KAMOTE system, based on its simulated testing and demonstration. It is composed of two parts: Part I evaluates the system using the FURPS model (Functionality, Usability, Reliability, Performance, and Supportability), and Part II evaluates the system's perceived usability using the System Usability Scale (SUS). Kindly answer both parts as honestly and objectively as possible, as your assessment will be used to validate and further improve the system.

Part I. FURPS Evaluation

Rate each item using the scale below:

4 = Highly Acceptable 3 = Acceptable 2 = Needs Revision 1 = Not Acceptable

Criteria	Rating
F – Functionality	
The system accurately detects helmet non-compliance among motorcycle riders.	

Criteria	Rating
The system accurately detects other defined violations (e.g., missing side mirror, counterflow driving, illegal turning, passenger overloading).	
The system correctly reads and records the license plate of the vehicle involved in a detected violation.	
The captured evidence (frame, plate number, and violation details) would be sufficient to support a non-contact citation, similar to existing no-contact apprehension programs.	
U – Usability	
The administrative dashboard is easy to navigate and understand.	
Violation logs and captured evidence frames are presented clearly and are easy to interpret.	
The information displayed (timestamp, plate number, violation type, and evidence frame) is sufficient for documentation purposes.	
R – Reliability	
The simulated violations and results presented are realistic and representative of actual road and traffic conditions.	
The system performs consistently across the different simulated lighting and traffic conditions shown.	
The rate of false detections (false positives or false negatives) observed is within an acceptable level.	
P – Performance	
The system processes video footage and produces results within a reasonable amount of time.	
S – Supportability	
The handling of captured footage (e.g., faces, plate numbers) shows adequate consideration for data privacy.	
Access to the dashboard and violation records appears properly restricted and secured.	
The system's logging and recording method appears suitable for integration into the office's enforcement workflow.	
Overall Acceptability	
Overall, the system shows strong potential as a tool for future automated, non-contact traffic enforcement in Zamboanga City.	

Overall Recommendation

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☐ Highly Acceptable for Enforcement Use ☐ Acceptable with Minor Revisions

☐ Acceptable with Major Revisions ☐ Not Acceptable

Part II. System Usability Scale (SUS)

For each statement below, please encircle or mark the number that best reflects how strongly you agree or disagree, using the 5-point scale from Strongly Disagree (1) to Strongly Agree (5). Kindly respond to every item, even if you are not entirely certain of your answer.

No.	Statement	1 (Strongly Disagree)	2	3	4	5 (Strongly Agree)
1	I think that I would like to use this system frequently.	○	○	○	○	○
2	I found the system unnecessarily complex.	○	○	○	○	○
3	I thought the system was easy to use.	○	○	○	○	○
4	I think that I would need the support of a technical person to be able to use this system.	○	○	○	○	○
5	I found the various functions in this system were well integrated.	○	○	○	○	○
6	I thought there was too much inconsistency in this system.	○	○	○	○	○
7	I would imagine that most people would learn to use this system very quickly.	○	○	○	○	○
8	I found the system very cumbersome to use.	○	○	○	○	○
9	I felt very confident using the system.	○	○	○	○	○
10	I needed to learn a lot of things before I could get going with this system.	○	○	○	○	○

Additional Comments/Suggestions:

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Evaluated by:

Name:
Signature:

Prepared By:

EMMAN NICHOLAS BLABE B. IDULSA
JEHANA H. KAIRAN
FATIMA SHEENA SHARIFA T. ABUBAKAR
Researchers

Noted By:

JAYDEE C. BALLAHO, MIT
Thesis Instructor
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MARK L. FLORES, PhD
Dean, College of Computing Studies

Appendix D: Relevant Source Code

Appendix E: Ethical Clearance

Appendix F: Plagiarism Report

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Appendix G: Research Critique and Editing Certificate

Appendix H: Curriculum Vitae

Note: 1 page per researcher